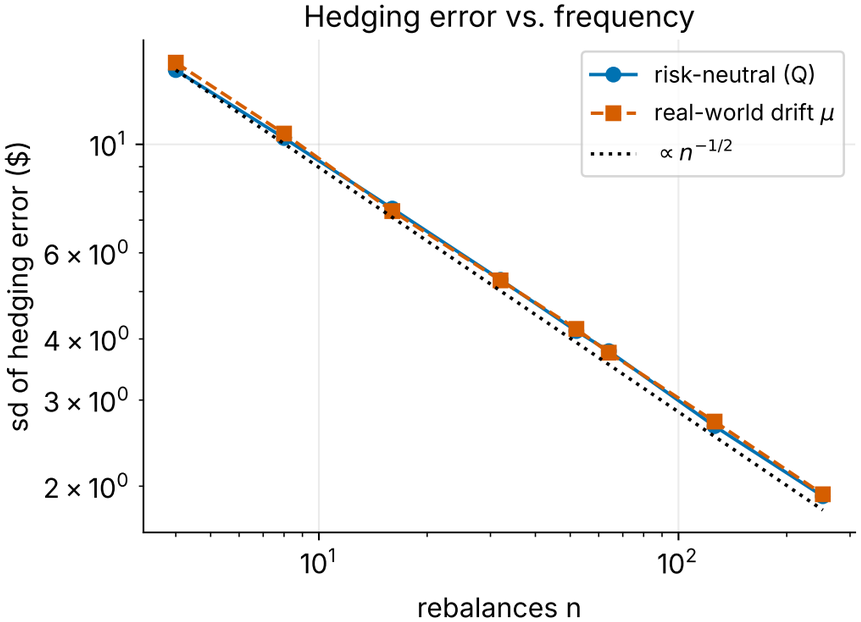
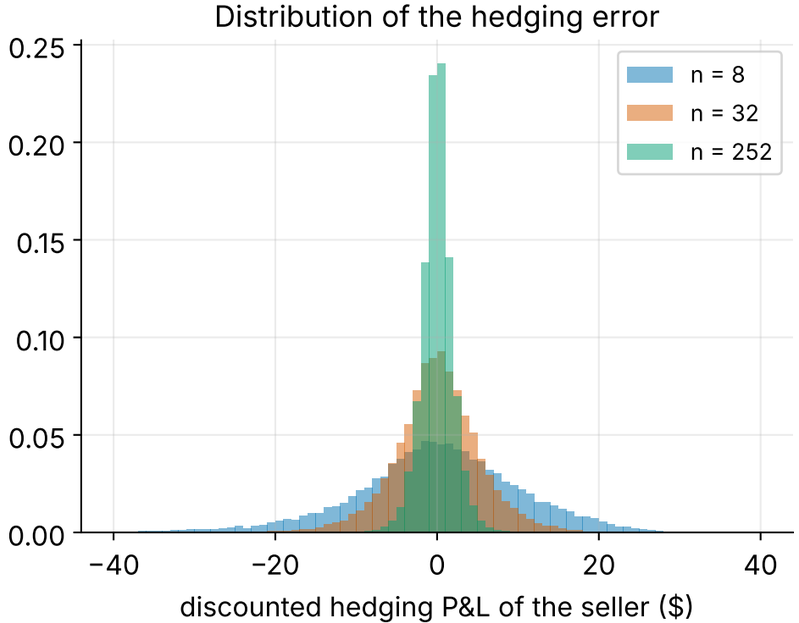
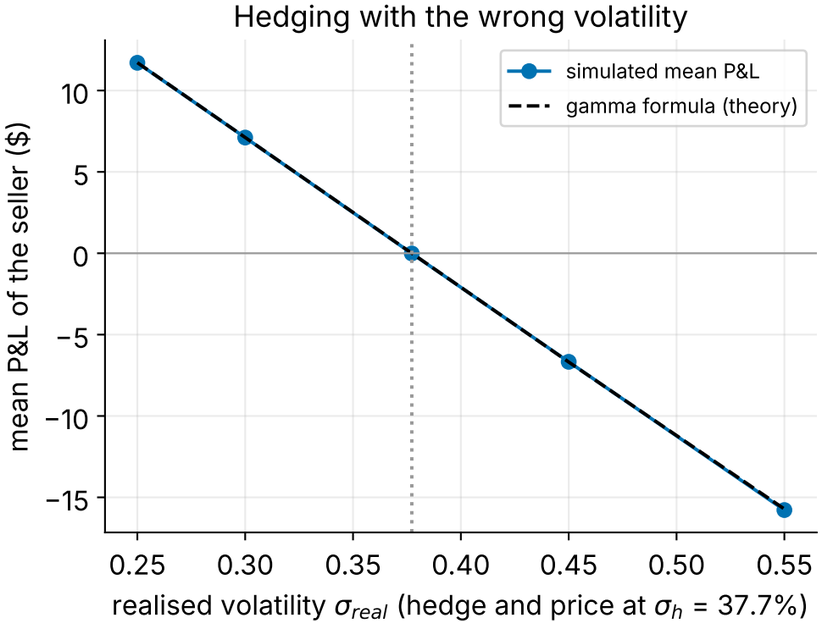

Hedging error vs. frequency
risk-neutral (Q)
real-world drift μ
$\propto n^{-1/2}$
sd of hedging error ($)
10^1
6×10^0
4×10^0
3×10^0
2×10^0
10^1
10^2
rebalances n


Distribution of the hedging error
n = 8
n = 32
n = 252
0.25
0.20
0.15
0.10
0.05
0.00
−40
−20
0
20
40
discounted hedging P&L of the seller ($)


Hedging with the wrong volatility
simulated mean P&L
gamma formula (theory)
mean P&L of the seller ($)
10
5
0
−5
−10
−15
0.25
0.30
0.35
0.40
0.45
0.50
0.55
realised volatility σ_{real} (hedge and price at σ_h = 37.7%)